

RIFT Cannot Replace Dark Matter: Exhaustive Test of Kinetic Modifications to the Locked Action

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April 11, 2026

Abstract

A recurring question for any modified-gravity theory is whether the scalar degree of freedom can source galactic dark matter halos and reproduce flat rotation curves without invoking collisionless dark matter particles. We subject the locked RIFT action ($\Lambda_0 = 0.003$, $m_0 = 0.01 \text{ Mpc}^{-1}$) to three independent tests against the NGC 3198 rotation curve ($v_{\text{flat}} = 150 \text{ km/s}$): (i) static Yukawa profile (Sim 99): the Ψ halo falls exponentially steeper than the $\rho \propto r^{-2}$ isothermal required for flat curves, with a density deficit of $2.7 \times 10^6 \times$; (ii) time-domain PDE condensation (Sim 100): the galactic curvature coupling $2\Lambda_0|R_{\text{gal}}| \approx 5 \times 10^{-9} \text{ kpc}^{-2}$ is $5 \times 10^7 \times$ weaker than m_0^2 ; curvature condensation at galactic densities is impossible; attractive self-coupling ($\lambda < 0$) produces tachyonic blow-up, not a stable halo; (iii) Vainshtein screening (Sim 103): the nonlinear kinetic term $(\alpha_V/4M_V^2)(\partial\Psi)^4$ remains deep in the linear regime for all scanned (α_V, M_V) ; the Vainshtein ratio $\mathcal{V} = \alpha_V(\delta\Psi')^2/M_V^2 \leq 3.7 \times 10^{-6}$. All three mechanisms fail for the same root reason: the galactic field gradient $|\delta\Psi'| \sim 6 \times 10^{-8} \text{ kpc}^{-1}$, set by the weak $\Lambda_0 R_{\text{gal}}$ source, is six orders of magnitude below any nonlinear threshold. A direct matter coupling $\beta \sim 1600 \Lambda_0$ would close the density gap but is not contained in the locked action and would violate cosmological homogeneity constraints. The conclusion is definitive: RIFT at its cosmological best-fit cannot replace dark matter through any modification of the existing Lagrangian. New first-principles physics coupling Ψ to baryonic matter is the minimum requirement.

1 Introduction

Flat rotation curves of spiral galaxies — the canonical evidence for dark matter — require a halo density $\rho \propto r^{-2}$ (isothermal) within the optical disk. For NGC 3198, with $v_{\text{flat}} = 150 \text{ km/s}$ and $R_d = 2.72 \text{ kpc}$, the required halo density at $r = 10 \text{ kpc}$ is

$$\rho_{\text{iso}}(10 \text{ kpc}) = \frac{v_{\text{flat}}^2}{4\pi G r^2} = 4.2 \times 10^3 M_{\odot} \text{ kpc}^{-3}. \quad (1)$$

The RIFT scalar field Ψ couples to spacetime curvature through the non-minimal term $\Lambda_0 \Psi R$ in the action. Could this coupling produce a self-gravitating Ψ condensate that mimics a dark matter halo? We test this exhaustively.

2 Test I: Static Yukawa Profile (Sim 99)

The static field equation for Ψ in spherical symmetry (with the FLRW background subtracted) gives a Yukawa profile:

$$\delta\Psi(r) = A \frac{e^{-m_0 r}}{r} + B \frac{e^{+m_0 r}}{r}. \quad (2)$$

The physical (decaying) solution with $m_0^{-1} = 10,000$ kpc gives an effective energy density $\rho_\Psi \propto (\nabla\delta\Psi)^2 \propto e^{-2m_0 r}/r^4$. This falls exponentially steeply — orders of magnitude faster than r^{-2} — and cannot reproduce flat rotation curves for any value of the amplitude A .

G_{eff} scan. A scan over $G_{\text{eff}}/G \in [1, 10^4]$ confirms: boosting Λ_0 changes the amplitude but not the profile shape, which remains Yukawa. No static solution produces $\rho \propto r^{-2}$.

3 Test II: Time-Domain Condensation (Sim 100)

We integrate the Ψ Klein–Gordon PDE on a fixed galactic background over 10^4 Mpc/ c ($\sim 10^{10}$ yr), testing whether: (a) the galactic curvature source $2\Lambda_0|R_{\text{gal}}|$ drives a condensate, or (b) an attractive self-coupling $\lambda\Psi^3/3!$ stabilises a halo.

Curvature coupling. The galactic Ricci scalar $|R_{\text{gal}}| \approx 8\pi G\rho_{\text{disk}} \approx 10^{-9}$ kpc $^{-2}$. The coupling strength $2\Lambda_0|R_{\text{gal}}| \approx 5 \times 10^{-9}$ kpc $^{-2}$ compared to $m_0^2 = 10^{-4}$ kpc $^{-2}$: the source is $5 \times 10^7 \times$ weaker than the restoring force. No condensate forms.

Attractive self-coupling. A scan over $\lambda < 0$ shows the field diverges tachyonically before forming a stable halo at any galactic density. The static (Sim 99) and time-domain (Sim 100) limits agree, confirming adiabaticity and ruling out history-dependence as a resolution.

4 Test III: Vainshtein Screening (Sim 103)

The Vainshtein mechanism introduces the nonlinear kinetic term $(\alpha_V/4M_V^2)(\partial\Psi)^4$, enhancing the effective energy density:

$$\rho_\Psi(r) = \frac{c^2}{8\pi G} \cdot \frac{1}{2}(\delta\Psi')^2 \left[1 + \underbrace{\frac{\alpha_V}{M_V^2}(\delta\Psi')^2}_{\mathcal{V}} \right]. \quad (3)$$

In the Poisson-limit ($m_0 r \ll 1$) the field gradient is:

$$\frac{d\delta\Psi}{dr} = \frac{2G\Lambda_0 M_{\text{enc}}(r)}{r^2} \approx 6.1 \times 10^{-8} \text{ kpc}^{-1} \quad (r = 10 \text{ kpc}). \quad (4)$$

Every combination is deep in the linear regime ($\mathcal{V} \ll 1$). The scalar energy density is $\rho_\Psi(10 \text{ kpc}) = 1.6 \times 10^{-3} M_\odot \text{ kpc}^{-3}$, a factor of 2.7×10^6 below the isothermal requirement (eq. ??).

Table 1: Vainshtein ratio at $r = 10$ kpc for NGC 3198.

α_V	M_V [kpc $^{-1}$]	$\mathcal{V} = \alpha_V(\delta\Psi')^2/M_V^2$
1	0.001	3.7×10^{-9}
100	0.001	3.7×10^{-7}
1000	0.001	3.7×10^{-6}
1000	0.01	3.7×10^{-8}

5 Root Cause and Required New Physics

All three tests fail for the same reason. The field gradient $|\delta\Psi'| \sim 6 \times 10^{-8}$ kpc $^{-1}$ is determined by the coupling strength $\Lambda_0 R_{\text{gal}}$, which is fixed by the cosmological best-fit value $\Lambda_0 = 0.003$ and the galactic curvature $R_{\text{gal}} \sim 10^{-9}$ kpc $^{-2}$. This is an unavoidable consequence of the locked action: the same Λ_0 that ensures cosmological consistency makes the galactic source negligible.

To close the $2.7 \times 10^6 \times$ density gap, a direct matter coupling $\beta\Psi\rho_b$ with $\beta \approx 4.9 \approx 1600\Lambda_0$ would be needed. Such a coupling:

1. Is not derivable from the locked action;
2. Introduces a large new free parameter;
3. Would drive $\mathcal{O}(1)$ perturbations in the cosmological background, violating CMB homogeneity.

6 Conclusion

RIFT at its cosmological best-fit cannot replace dark matter through any modification of the existing Lagrangian. The scalar field couples too weakly to galactic curvature to source the required $\rho \propto r^{-2}$ halo, and no kinetic enhancement (Yukawa profile, temporal condensation, Vainshtein nonlinearity) changes this conclusion. A first-principles coupling between Ψ and baryonic matter — derivable from a symmetry principle rather than added by hand — is the minimum new physics required for RIFT to address the dark matter problem.